

```

import java.util.Scanner;

public class SeriesSum {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Number of cases: ");

        int cases = sc.nextInt();

        for (int i = 1; i <= cases; i++) {

            System.out.print("Case " + i + ": Enter n" + i + ": ");

            int n = sc.nextInt();

            // Series and sum output in one loop

            int sum = 0;

            for (int j = 1; j <= n; j++) {

                System.out.print(j + (j < n ? "+" : ""));

                sum += j;

            }

            System.out.println(" = " + sum);

        }

        sc.close();

    }

}

```

```

Number of cases: 3
Case 1: Enter n1: 3
1+2+3 = 6
Case 2: Enter n2: 4
1+2+3+4 = 10
Case 3: Enter n3: 5
1+2+3+4+5 = 15

```